

VI. LINUX PRIVILEGE ESCALATION

Aim:

Learn the fundamentals of Linux privilege escalation. From enumeration to exploitation, get hands-on with over 8 different privilege escalation techniques

Procedure:

1. Log into the target system and identify the current user.
2. Gather system information such as OS version and kernel details.
3. Check user permissions and search for SUID/SGID files.
4. Look for misconfigurations like writable files, cron jobs, or stored credentials.
5. Identify and exploit vulnerabilities (e.g., sudo permissions or vulnerable binaries).
6. Escalate privileges to root and verify access.

Tasks:

Task 1: Introduction

Read the above.

No answer needed

✓ Correct Answer

Task 2: What is Privilege Escalation?

Read the above.

No answer needed

✓ Correct Answer

Task 3: Enumeration

What is the hostname of the target system?

wade7363

✓ Correct Answer

What is the Linux kernel version of the target system?

3.13.0-24-generic

✓ Correct Answer

What Linux is this?

Ubuntu 14.04 LTS

✓ Correct Answer

What version of the Python language is installed on the system?

2.7.6

✓ Correct Answer

What vulnerability seem to affect the kernel of the target system? (Enter a CVE number)

CVE-2015-1328

✓ Correct Answer

Task 4: Automated Enumeration Tools

Install and try a few automated enumeration tools on your local Linux distribution

No answer needed

✓ Correct Answer

Task 5: Privilege Escalation: Kernel Exploits

find and use the appropriate kernel exploit to gain root privileges on the target system.

No answer needed

✓ Correct Answer



What is the content of the flag1.txt file?

THM-28392872729920

✓ Correct Answer

Task 6: Privilege Escalation: Sudo

How many programs can the user "karen" run on the target system with sudo rights?

3

✓ Correct Answer

What is the content of the flag2.txt file?

THM-402028394

✓ Correct Answer

How would you use Nmap to spawn a root shell if your user had sudo rights on nmap?

sudo nmap --interactive

✓ Correct Answer

What is the hash of frank's password?

\$6\$2.sUUDsOLipXKxcr\$elmtgFExyr2ls4jsghdD3DHLHHP9X50lv.jNmwo/BJpphrPRJWjelWEz2HH.joV14aDEwW1c3CahzB1uaq6

✓ Correct Answer

Task 7: Privilege Escalation: Kernel SUID

Which user shares the name of a great comic book writer?

gerryconway

✓ Correct Answer

What is the password of user2?

Password1

✓ Correct Answer

What is the content of the flag3.txt file?

THM-3847834

✓ Correct Answer

Task 8: Privilege Escalation: Capabilities

Complete the task described above on the target system

No answer needed

✓ Correct Answer

How many binaries have set capabilities?

6

✓ Correct Answer

What other binary can be used through its capabilities?

view

✓ Correct Answer

What is the content of the flag4.txt file?

THM-9349843

✓ Correct Answer

Task 9: Privilege Escalation: Cron Jobs

How many user-defined cron jobs can you see on the target system?

4

✓ Correct Answer

What is the content of the flag5.txt file?

THM-383000283

✓ Correct Answer

What is Matt's password?

123456

✓ Correct Answer

Task 10: Privilege Escalation: Path

What is the odd folder you have write access for?

/home/murdoch

✓ Correct Answer



Exploit the \$PATH vulnerability to read the content of the flag6.txt file.

No answer needed

✓ Correct Answer



What is the content of the flag6.txt file?

THM-736628929

✓ Correct Answer

Task 11: Privilege Escalation: NFS

How many mountable shares can you identify on the target system?

3

✓ Correct Answer

How many shares have the "no_root_squash" option enabled?

3

✓ Correct Answer

Gain a root shell on the target system

No answer needed

✓ Correct Answer

What is the content of the flag7.txt file?

THM-89384012

✓ Correct Answer

Task 12: Capstone Challenge

What is the content of the flag1.txt file?

THM-42828719920544

✓ Correct Answer

What is the content of the flag2.txt file?

THM-168824782390238

✓ Correct Answer

Result:

Successfully performed Linux privilege escalation by identifying system misconfigurations and exploiting available vulnerabilities, thereby gaining root access to the target system.